

Cryptographic Risk Summary

Scan **cf02f22e** · generated 03 September 2026, 20:34 UTC · sources: source, dependencies, binary, tls

319	151	25	141	86	144
ASSETS	BROKEN	DEPRECATED	ADEQUATE	HARVEST-NOW	BANNED 2035

Scope: source:targets, source:/usr/bin/ssh, source:/usr/bin/curl, source:/usr/bin/git, dependencies:targets, dependencies:/usr/bin/ssh, dependencies:/usr/bin/curl, dependencies:/usr/bin/g

Basis for these numbers

Assets are scored against two independent clocks. The **quantum** clock uses Mosca's inequality with a quantum-arrival estimate of **Z = 2031**, which is an input set by the risk owner and not a prediction this tool makes. The **regulatory** clock uses NIST IR 8547, under which RSA and elliptic-curve cryptography are deprecated after 2030 and disallowed after 2035; those dates are fixed and do not depend on any Q-Day estimate.

Of 319 assets, **71** have scores that move if Z changes. The remaining **248** are pinned by the regulatory deadline or by an outright cryptographic break, and their priority is therefore independent of any view about when quantum computers arrive.

Risk distribution

	LOW URGENCY	MEDIUM URGENCY	HIGH URGENCY
HIGH CRITICALITY	33	1	57
MEDIUM CRITICALITY	196	28	4
LOW CRITICALITY	0	0	0

Migration priority

The 40 highest-scoring assets, ordered by Mosca exposure and regulatory deadline weighted by business criticality. A **!** marks data whose confidentiality requirement outlives the estimated time to a quantum computer, so recorded traffic is exposed today.

SCORE	ASSET	STATUS	CRIT	CONF	LOCATION	RECOMMENDED ACTION
1.00 !	ECDH-x25519	Broken	high	high	paramiko/paramiko/kex_curve25519.py:45	FIPS 203 ML-KEM-768
1.00 !	ECDH-x25519	Broken	high	high	paramiko/paramiko/kex_curve25519.py:30	FIPS 203 ML-KEM-768
1.00 !	ECDH-x25519	Broken	high	high	paramiko/paramiko/kex_mlkem.py:84	FIPS 203 ML-KEM-768
1.00 !	ECDH-x25519	Broken	high	high	paramiko/paramiko/kex_mlkem.py:76	FIPS 203 ML-KEM-768
1.00 !	RSA	Broken	high	high	paramiko/paramiko/rsa.py:139	FIPS 203 ML-KEM-768

SCORE	ASSET	STATUS	CRIT	CONF	LOCATION	RECOMMENDED ACTION
1.00 !	RSA	Broken	high	high	paramiko/paramiko/rsakey.py:167	FIPS 204 ML-DSA-65
1.00 !	ECDSA	Broken	high	medium	paramiko/paramiko/ecdsakey.py:268	FIPS 204 ML-DSA-65
1.00 !	ECDSA	Broken	high	medium	paramiko/paramiko/kex_ecdh_nist.py:59	FIPS 204 ML-DSA-65
1.00 !	RSA	Broken	high	medium	paramiko/paramiko/rsakey.py:184	FIPS 203 ML-KEM-768
0.80 !	SHA-1	Deprecated / insufficient	high	high	paramiko/paramiko/config.py:449	FIPS 197 / FIPS 180-4 SHA-256
0.80 !	SHA-1	Deprecated / insufficient	high	high	paramiko/paramiko/hostkeys.py:299	FIPS 197 / FIPS 180-4 SHA-256
0.80 !	SHA-1	Deprecated / insufficient	high	high	paramiko/paramiko/hostkeys.py:294	FIPS 197 / FIPS 180-4 SHA-256
0.70 !	ECDH	Broken	high	medium	jjwt/api/src/main/java/io/jsonwebtoken/security/Jwks.java:243	FIPS 204 ML-DSA-65
0.70 !	ECDH	Broken	high	medium	jjwt/api/src/main/java/io/jsonwebtoken/security/Jwks.java:256	FIPS 204 ML-DSA-65
0.70 !	ECDH	Broken	high	medium	jjwt/impl/src/main/java/io/jsonwebtoken/impl/security/EdwardsC	FIPS 204 ML-DSA-65
0.70 !	ECDH	Broken	high	medium	jjwt/impl/src/main/java/io/jsonwebtoken/impl/security/EdwardsC	FIPS 204 ML-DSA-65
0.70 !	ECDH	Broken	high	medium	jjwt/impl/src/main/java/io/jsonwebtoken/impl/security/EcdhKeyA	FIPS 204 ML-DSA-65
0.70 !	ECDH	Broken	high	medium	jjwt/impl/src/main/java/io/jsonwebtoken/impl/security/JcaTempl	FIPS 204 ML-DSA-65
0.70 !	ECDSA	Broken	high	medium	jjwt/api/src/main/java/io/jsonwebtoken/SignatureAlgorithm.java	FIPS 204 ML-DSA-65
0.70 !	ECDSA	Broken	high	medium	jjwt/api/src/main/java/io/jsonwebtoken/SignatureAlgorithm.java	FIPS 204 ML-DSA-65
0.70 !	ECDSA	Broken	high	medium	jjwt/api/src/main/java/io/jsonwebtoken/SignatureAlgorithm.java	FIPS 204 ML-DSA-65
0.70 !	ECDSA	Broken	high	medium	jjwt/api/src/main/java/io/jsonwebtoken/SignatureAlgorithm.java	FIPS 204 ML-DSA-65
0.70 !	ECDSA	Broken	high	medium	jjwt/api/src/main/java/io/jsonwebtoken/Jwts.java:269	FIPS 204 ML-DSA-65
0.70 !	ECDSA	Broken	high	medium	jjwt/api/src/main/java/io/jsonwebtoken/Jwts.java:276	FIPS 204 ML-DSA-65

SCORE	ASSET	STATUS	CRIT	CONF	LOCATION	RECOMMENDED ACTION
0.70 !	ECDSA	Broken	high	medium	jjwt/api/src/main/java/io/jsonwebtoken/Jwts.java:283	FIPS 204 ML-DSA-65
0.70 !	ECDSA	Broken	high	medium	jjwt/impl/src/main/java/io/jsonwebtoken/impl/security/Provided	FIPS 204 ML-DSA-65
0.70 !	ECDSA	Broken	high	medium	jjwt/impl/src/main/java/io/jsonwebtoken/impl/security/ECCurve.	FIPS 204 ML-DSA-65
0.70 !	ECDSA	Broken	high	medium	jjwt/impl/src/main/java/io/jsonwebtoken/impl/security/DefaultE	FIPS 204 ML-DSA-65
0.70 !	ECDSA	Broken	high	medium	jjwt/impl/src/main/java/io/jsonwebtoken/impl/security/EcSignat	FIPS 204 ML-DSA-65
0.70 !	ECDSA	Broken	high	medium	jjwt/impl/src/main/java/io/jsonwebtoken/impl/security/EcSignat	FIPS 204 ML-DSA-65
0.70 !	ECDSA	Broken	high	medium	jjwt/impl/src/main/java/io/jsonwebtoken/impl/security/JwkConve	FIPS 204 ML-DSA-65
0.70 !	EdDSA	Broken	high	medium	jjwt/api/src/main/java/io/jsonwebtoken/Jwts.java:304	FIPS 204 ML-DSA-65
0.70 !	EdDSA	Broken	high	medium	jjwt/api/src/main/java/io/jsonwebtoken/security/Jwks.java:217	FIPS 204 ML-DSA-65
0.70 !	EdDSA	Broken	high	medium	jjwt/api/src/main/java/io/jsonwebtoken/security/Jwks.java:230	FIPS 204 ML-DSA-65
0.70 !	EdDSA	Broken	high	medium	jjwt/impl/src/main/java/io/jsonwebtoken/impl/security/EdwardsC	FIPS 204 ML-DSA-65
0.70 !	EdDSA	Broken	high	medium	jjwt/impl/src/main/java/io/jsonwebtoken/impl/security/EdwardsC	FIPS 204 ML-DSA-65
0.70 !	EdDSA	Broken	high	medium	jjwt/impl/src/main/java/io/jsonwebtoken/impl/security/EdwardsC	FIPS 204 ML-DSA-65
0.70 !	EdDSA	Broken	high	medium	jjwt/impl/src/main/java/io/jsonwebtoken/impl/security/EdSignat	FIPS 204 ML-DSA-65
0.70 !	RSA	Broken	high	medium	jjwt/api/src/main/java/io/jsonwebtoken/SignatureAlgorithm.java	FIPS 204 ML-DSA-65
0.70 !	RSA	Broken	high	medium	jjwt/api/src/main/java/io/jsonwebtoken/SignatureAlgorithm.java	FIPS 204 ML-DSA-65

Assumptions and limitations

Untagged business inputs. 27 of 319 assets were scored using a default data lifetime because the owning service carries no data-classification tag. No scanner can determine how long data must stay confidential; that is a business fact. Tagging those services will change their scores.

Detection confidence. 0 findings are low confidence, meaning the evidence establishes that an algorithm is present but not that the code path is reached. Findings are labelled individually rather than presented as uniformly certain.

What static analysis cannot see. Algorithms selected at runtime from configuration, key sizes read from variables, and indirect API resolution are not detected. Measured recall against the project's labelled corpus is published with the source; this report does not claim completeness.

Symmetric cryptography. AES-128 and SHA-256 are reported as adequate. Grover's algorithm gives only a quadratic speedup, requires infeasible sequential quantum computation, and parallelises poorly; NIST defines its own post-quantum security categories using these algorithms as reference points. Where a compliance regime requires AES-256 or SHA-384 that is reported as a policy obligation, not as a cryptographic weakness.

Standard conformance

The full inventory is exportable as a CycloneDX 1.7 Cryptography Bill of Materials, ratified as ECMA-424 2nd Edition, suitable for submission as regulatory evidence. This PDF is a summary of that inventory, not a replacement for it.